

LLM-as-a-Judge Report: Repair-understanding ARC

What was measured?

The judge scores how much the model loses the intended meaning of the question because of corrupted wording. A score of 0 means the meaning is fully understood; 5 means the model is fundamentally lost. This is a semantic-understanding score, not a direct correctness label. The judge returns the score, a verbatim evidence quote when available, and a one-line summary. The tables below use 0–5 scores and include only valid answered traces from `arc_judge_traces.jsonl`.

Examples at different scores

The examples below show the score, final-answer outcome, the judge’s evidence, and its one-line summary. An empty evidence field means that the trace contained no suitable verbatim cue for that dimension.

Trace `typo25_real10`, index 0. **Score:** 0 on the 0–5 scale. **Final-answer outcome:** correct.
Judge evidence:

The question is asking which effect is most likely due to this increase in rotation

One-line summary: The trace shows a straightforward analysis with brief moments of reconsideration, leading to a confident conclusion

Trace `typo75_real70`, index 0. **Score:** 2 on the 0–5 scale. **Final-answer outcome:** correct.
Judge evidence:

Okay, so I have this question here about an astronomer observing a planet rotating faster after a meteorite impact.

One-line summary: The trace exhibits repeated reconsideration and second-guessing, particularly when evaluating the effects of increased rotation on planetary gravity and days, leading to a score of 6 for self-doubt.

Trace `typo75_real40`, index 32. **Score:** 5 on the 0–5 scale. **Final-answer outcome:** correct.
Judge evidence:

The question is about a type of hat. Hmm, hats are made of materials like wool, silk, maybe leather. But the options are talking about ice and air. That seems a bit confusing.

One-line summary: The model’s trace is filled with repeated reconsiderations and second-guessing, as it struggles to reconcile the question about hats with options discussing ice and air.

Aggregate tables

Table 1: Mean score by real-word typo ratio

real_ratio	n	repair_understanding_score_mean
10	1442	0.938
40	1412	1.099
70	1447	1.314

Table 2: Mean score by typo percentage

typo_percentage	n	repair_understanding_score_mean
0	443	0.029
25	1448	0.969
50	1415	1.028
75	1438	1.355

Table 3: Mean score by final-answer correctness

outcome	n	repair_understanding_score_mean
correct	3888	0.883
wrong	856	1.617

Consequences

In this run, more typos are associated with less ability to recover and understand the question meaning: the mean rises from 0.053 for clean traces to 0.580 at 25 percent typos, 0.576 at 50 percent, and 0.764 at 75 percent. Higher scores mean more loss of the intended meaning.